Reading Folded Hands: Opponent Modelling in Heads-Up Limit Hold'em

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Abstract

To beat a poker opponent you need to know how they play, but you only see their cards when a hand is played to the end. We model the opponent with a *Bayesian network*: their hidden cards and the public cards decide how strong their hand is, and strength, the situation at the table and their current playing style decide what they do. After every action the network gives a probability for each pair of cards they might hold. The network also shows that hands where the cards stay hidden can still be used for learning, while learning only from shown hands is biased. On 120,000 logged hands between poker programs, where every card is known, this makes the learned behaviour about 80 times more accurate and improves decisions. On 87,428 hands between humans, it finds that people change style about every 50 to 65 hands, while programs almost never do.

1 Introduction

In heads-up limit Texas Hold'em, two players each get two private cards and share five public cards, dealt over three rounds (flop, turn, river) with betting before and after each. A hand ends when someone *folds*, and no cards are shown, or at the *showdown*, where both hands are shown. The best programs already play close to an unbeatable equilibrium [3]. To win more, a player has to exploit how one particular opponent deviates from it, and learn this mostly from hands where the opponent's cards are never seen.

Bayesian networks have been used for this since the Bayesian Poker Player [8, 11]. Most such models learn only from showdowns [8, 11, 15]; models that also use folded hands [12, 13] assume an opponent who never changes; and changing opponents have been tracked only in tiny games [1]. Our network combines both: it reads the cards through all rounds of a hand, and lets the opponent's style change between hands. Because competition logs record every card [7], we can check it against the truth. We show that

- learning only from showdowns is badly biased, and using folded hands fixes it (Sections 3 and 4);
- one model per opponent reads cards better and makes better river decisions, but does not yet win more over whole hands;
- the network detects a change of opponent within about ten hands, and humans change style far more often than programs.

2 The model

We describe one opponent from the seat of the other player, the *observer*, using the probability vocabulary of [2]. A *session* is the sequence of hands against the same opponent. Figure 1 shows the network.

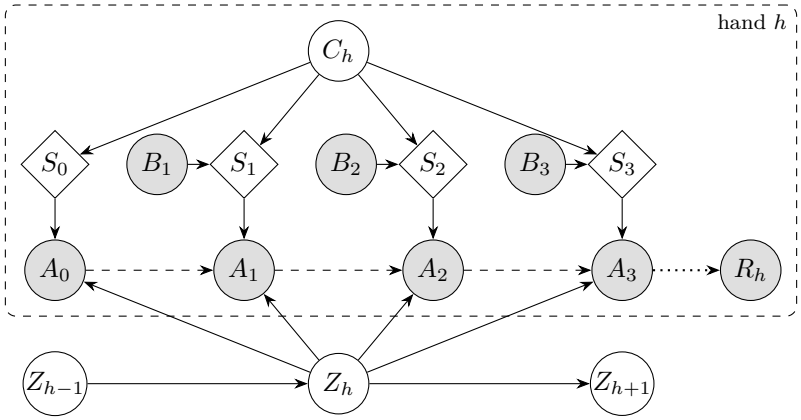


Figure 1: The network. Inside a hand (dashed box), the opponent's cards C and the board B give a strength S for each round, and the strength drives the actions A. The style Z also drives the actions and can change from hand to hand. Grey: what the observer sees. Diamonds: computed, not random. Whether the cards are shown (R) is decided by the actions alone.

What is in a hand. The opponent holds a pair of cards C, one of 1326 possible pairs. Before any betting, every pair that does not clash with a card we can see is equally likely. On each round the pair has a *strength* S from 1 (weakest) to 8 (strongest): we rank all pairs by how likely they are to win on the current board [5] and cut the ranking into eight equal groups. Strength is recomputed every round, so a pair can be weak on the flop and strong on the river; this is what ties the rounds together. The opponent's action A is fold, check/call or bet/raise, taken in a *situation* X: the round, whether they face a bet, how many bets have been made, and their position (80 situations in all).

How the opponent acts. The opponent's behaviour is a table

$$\theta_{z,x,s}(a) = P(\text{action } a \mid \text{style } z,\ \text{situation } x,\ \text{strength } s),$$

for example "in this situation, with a strength-2 hand, this opponent bets 10% of the time". The *style* Z is hidden; it stands for how the opponent is playing right now, say cautiously or aggressively. From one hand to the next the style stays the same with probability $1 - \varepsilon$ and otherwise switches, so the styles form a Markov chain.

Reading the cards. Bayes' rule gives the probability of each pair c after the actions seen so far:

$$P(c \mid \text{actions}) \ \propto \prod_{\text{actions}} \theta_{z,x,\,\text{strength of } c}(a).$$

In words: a pair becomes more likely if it would have made the observed actions likely. If the opponent raised on the river, strong pairs gain and weak pairs lose. We do this exactly for all 1326 pairs after every action. The style is tracked in the same spirit, hand by hand: each style gains weight when it explains the hand well.

Decisions. From the card probabilities we get q, our chance of holding the better hand. If the opponent bets b into a pot of P, calling risks b to win P, so calling pays off on average exactly when $q > b/(P + b)$.

Explaining a decision. Since everything is a probability, the network can say why it recommends an action, in three ways.

- *How sure?* The table θ is learned from a limited number of hands, so it is uncertain. We redraw plausible tables many times and recompute the decision; the *certificate* is the share of draws that agree. Near 100% the advice is safe; near 50% it is a coin flip.
- *Which action mattered?* Each opponent action gets a score: the log of how much more likely it is if the opponent is strong than if they are weak. The scores add up exactly to the total change in belief, so they show which action moved it most.
- *What would change my mind?* We swap one opponent action at a time and report the smallest change that flips the advice, e.g. "had they raised before the flop instead of calling, call".

3 Learning and data

Folded hands can be used. When a hand ends in a fold we do not see the opponent's cards, so we do not guess them: we add up the probability of the observed actions over every pair they could have held. This is safe for a simple reason. The cards stay hidden *because* of a fold, and the fold itself is seen. So the fact that the cards are hidden tells us nothing beyond the actions we already have, and averaging over the possible pairs loses no correctness.

Proposition 1. *If whether the cards are shown depends only on the observed actions, then learning from all hands, with the hidden pairs averaged out, is not biased by the missing cards.*

Learning only from shown hands is biased. A fold always ends the hand, so a fold never reaches the showdown: a model trained on showdowns never sees one and concludes that the opponent never folds. It also overrates every action that tends to keep hands going.

Learning the table. If all cards were known, θ would simply be counted: how often the opponent took each action with each strength in each situation. Since strengths and styles are partly hidden, we use the expectation–maximisation (EM) algorithm [4], which starts from a rough guess and repeats two steps until nothing changes:

1. *Count with fractions.* A shown hand adds a count of 1 to its true strength. A folded hand splits its count among the strengths it could have had, in proportion to their current probabilities. Styles are split the same way.
2. *Recompute the table* from these counts as frequencies.

To each count we add a few imaginary observations, spread like the opponent's overall action frequencies in that situation (which need no cards). This keeps rare situations from getting zero probabilities; their number λ sets how much the table is pulled towards these frequencies. A model of one opponent starts from the model of all opponents instead, so opponents seen in few hands borrow from the rest.

Data. We use 40 matches of 3000 hands from the 2011 Annual Computer Poker Competition (ACPC) [6, 7]: 120,000 hands whose logs show both players' cards, even after a fold. Each hand is used twice, once with each program as the opponent. The cards of folded hands are used only to score the models and for an *oracle*, a model counted from all true cards, which is the best our features allow. For human play we use the IRC Poker Database [10]: 87,428 two-player hands between 2,714 people, in 1,072 sessions. There only shown cards are known.

Fair testing. Each session is split in time: the first 60% of its hands for training, the next 10% for choosing settings, the last 30% for the final test, which is used once. The settings chosen were $\lambda = 2$, four styles (the most we tried, so more might help) and $\lambda = 1$ for single opponents.

4 Results

Results are on test hands; ± marks two standard deviations of the mean.

Showdown-only learning is biased. Table 1 shows how the first player acts before the flop, by hand quality. Over all hands, the weakest hands are folded a quarter of the time. Among showdown hands they are never folded, and they are raised more often (58% instead of 47%).

hand quality (weakest → strongest)	1	2	3	4	5
fold, all hands	25%	4%	3%	2%	1%
fold, showdown hands	0%	0%	0%	0%	0%
raise, all hands	47%	68%	68%	70%	77%
raise, showdown hands	58%	68%	68%	70%	75%

Table 1: First action before the flop, learned from all hands and from showdown hands only (all 120,000 hands).

The bias carries over to the whole table. Measured by the usual distance between probability tables (Kullback–Leibler divergence, 0 when identical), the showdown-only table is 0.475 from the oracle; EM with folded hands is 0.006, about 80 times closer.

Reading the cards and deciding. Table 2 gives two scores per model. Card reading is shown as an *effective number of pairs*: the model is as unsure as if it had narrowed the opponent to this many equally likely pairs (lower is better; e to the power of the average $-\log p(\text{true pair})$). Decisions are river calls and folds, scored with the real cards over 17,192 cases.

model	pairs left on the river	chips per decision vs. showdown-only
no information	990	–
only the last action [15]	909	–
showdown-only	780	0
EM, one style	738	-0.3 ± 0.4
EM, four styles	622	$+1.0 \pm 0.6$
EM, one model per opponent	573	$+1.4 \pm 0.7$
oracle per opponent (best possible)	550	$+1.6 \pm 0.7$
knowing the cards	1	$+11.1 \pm 0.7$

Table 2: Card reading and river decisions on test hands.

Three things stand out. Using folded hands and the full history beats showdown-only learning and beats looking at the last action only. Modelling each opponent separately helps most, close to the best possible with our features. And better card reading turns into better decisions only when the model is specific enough: one model for everybody does not gain. Even the best model captures a small part of what knowing the cards would give. The win probabilities are well calibrated (of the calls made at $q \approx 0.3$, about 30% win), and in Kuhn poker [9], a three-card game where the truth is known, advice certified at 80%, 90% and 95% was right 93%, 96% and 98% of the time.

Spotting a new opponent. Programs never change style during a match. To test the style chain on real play, we glue together test hands of two programs, switching about every 150 hands (Figure 2). Over 375 switches, the network is sure of the new program (above 90%) after a median of 11 hands and names the right program 79% of the time. Without using cards it is right 72% of the time; comparing the last 50 hands, 59%.

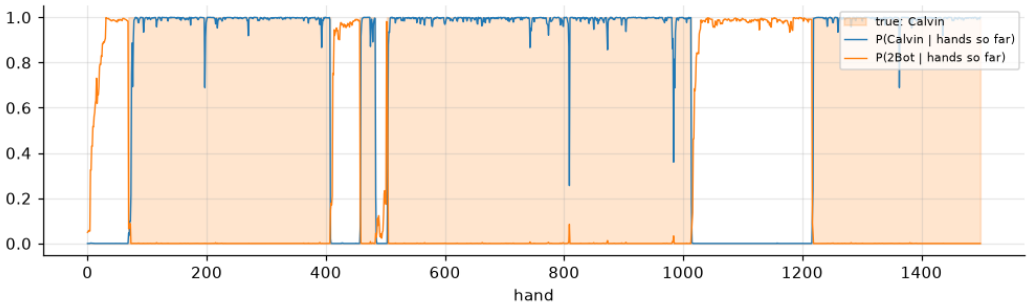


Figure 2: Two programs glued into one sequence. Shaded: the second program is playing. Lines: the network's probability for each program.

Playing whole hands. We also let the model play complete hands with a simple rule (fold below the pot odds, bet or raise when q is high), against copies of two ACPC programs: an aggressive one that bets or raises 61% of the time, and a passive one that only checks and calls. Here the results are not in the model's favour. Against the passive copy all agents tie. Against the aggressive one, an agent that ignores the opponent's actions wins 22.7 chips per hand, and nothing beats it: the model of all opponents loses 11.8 ± 2.9 chips per hand relative to it, because it reads the frequent bets as strength and folds too often; learning online during the match recovers most of this (-1.9 ± 1.6); the opponent's own model ties (-0.3 ± 1.0). The same pattern appears in the PettingZoo environment [14]. The lesson is that a model of the wrong opponent is expensive, and that learning on the fly limits the damage. These tests are small and the betting rule is crude.

Humans change style. On the IRC hands, models with two or three styles predict test hands better than one style. The learned switch chance is about 0.02 per hand, a change every 50 to 65 hands, against one every 700 hands for the programs. The styles differ mostly in aggression (23%, 31% and 64% bets and raises). A big loss raises the switch chance by only 0.005 ± 0.005, so we cannot say that losing makes people change style.

5 Conclusion

Modelling an opponent as a Bayesian network over cards, strength, actions and style lets every hand count, including the many where the cards stay hidden. Ignoring those hands is a real mistake: the learned behaviour ends up about 80 times further from the truth. With folded hands included, the network reads cards better, makes better river decisions when it is fitted to one opponent, explains its advice, spots a new opponent within about ten hands, and finds that humans change style often.

It does not yet win more over whole hands, and a model of the wrong opponent can lose heavily. The strength groups are coarse, and the decision gain is far from what perfect information would give. All checks need logged cards; in human play a loser may hide their cards, which breaks Proposition 1 unless showing is modelled too. Next steps are a smarter betting rule, finer strength features, and human data with full card logs.

References

[1] Nolan Bard and Michael Bowling. Particle filtering for dynamic agent modelling in simplified poker. In *Proc. AAAI*, 2007.

[2] Dimitri P. Bertsekas and John N. Tsitsiklis. Introduction to probability, 2000. Lecture notes, Course 6.041–6.431, M.I.T., Fall 2000.

[3] Michael Bowling, Neil Burch, Michael Johanson, and Oskari Tammelin. Heads-up limit hold'em poker is solved. *Science*, 347(6218):145–149, 2015.

[4] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society B*, 39(1):1–38, 1977.

[5] Michael Johanson, Neil Burch, Richard Valenzano, and Michael Bowling. Evaluating state-space abstractions in extensive-form games. In *Proc. AAMAS*, 2013.

[6] Juho Kim. Poker hand history file format specification. In *IEEE Conference on Games (CoG)*, 2024.

[7] Juho Kim et al. A dataset of poker hand histories. Zenodo, https://doi.org/10.5281/zenodo.10796885, 2025.

[8] Kevin B. Korb, Ann E. Nicholson, and Nathalie Jitnah. Bayesian poker. In *Proc. 15th Conference on Uncertainty in Artificial Intelligence (UAI)*, pages 343–350, 1999.

[9] Harold W. Kuhn. A simplified two-person poker. In *Contributions to the Theory of Games*, volume 1, pages 97–103. Princeton University Press, 1950.

[10] Michael Maurer. IRC poker database, 2001. University of Alberta Computer Poker Research Group, https://poker.cs.ualberta.ca/irc_poker_database.html.

[11] Ann E. Nicholson, Kevin B. Korb, and Darren Boulton. Using Bayesian decision networks to play Texas Hold'em poker. Technical report, Monash University, 2006.

[12] Marc Ponsen, Jan Ramon, Tom Croonenborghs, Kurt Driessens, and Karl Tuyls. Bayes-relational learning of opponent models from incomplete information in no-limit poker. In *Proc. AAAI*, 2008.

[13] Finnegan Southey, Michael Bowling, Bryce Larson, Carmelo Piccione, Neil Burch, Darse Billings, and Chris Rayner. Bayes' bluff: Opponent modelling in poker. In *Proc. 21st Conference on Uncertainty in Artificial Intelligence (UAI)*, 2005.

[14] J. K. Terry, Benjamin Black, Nathaniel Grammel, Mario Jayakumar, Ananth Hari, Ryan Sullivan, Luis Santos, Clemens Dieffendahl, Caroline Horsch, Rodrigo Perez-Vicente, Niall Williams, Yashas Lokesh, and Praveen Ravi. PettingZoo: Gym for multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

[15] Michael Terry and Brian Mihok. A Bayesian net inference tool for hidden state in Texas Hold'em poker. MIT 16.412J Final Project, https://web.mit.edu/16.412j/www/html/Final%20Projects/FIN_Terry_Mihok.pdf, 2005.